

Thesis A Seminar Summary

Multi-Channel Solar Car MPPT

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Thesis Aim

This thesis presents a **multi-channel MPPT system** for Sunswift 8, designed to **maximise energy extraction** under **dynamic** conditions.

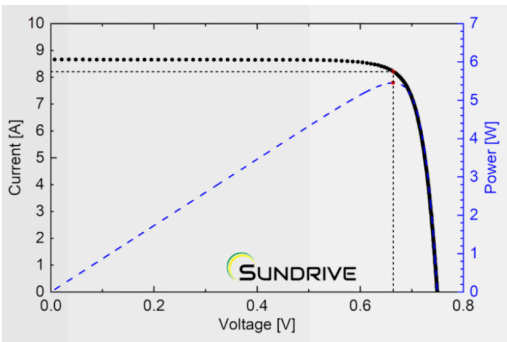
The system delivers power to both LV and HV buses depending on state-of-charge, and acts as a **flexible platform** for MPPT algorithm **development and validation**.

SR8 System Context & Requirements

- 5 solar arrays (~769 W) feeding LV (~48V) and HV (~690V) bus.
- Highly dynamic operating conditions during driving.
- Strict size, protection, & environmental constraints.
- Energy-limited system, efficiency is critical for range goals.

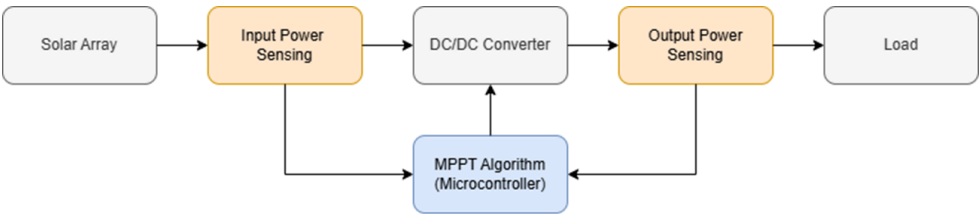
Array	Cells	V_{OC} [V]	I_{SC} [V]	V_{MPP} [V]	I_{MPP} [V]	P_{MPP} [W]
F1	41	30.50	8.66	27.59	8.22	226.81
BL	24	17.86	8.66	16.15	8.22	132.77
BR	24	17.86	8.66	16.15	8.22	132.77
R1	32	23.81	8.66	21.54	8.22	177.03
R2	18	13.39	8.66	12.11	8.22	99.58

General Background



IV/PV Curve for SR8 Cells

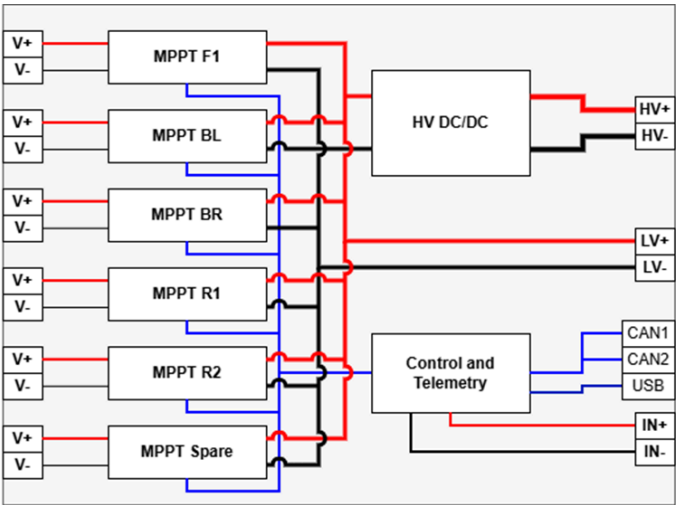
- Solar arrays exhibit non-linear I-V characteristics with a maximum power point (MPP).
- MPP varies with irradiance, temperature, and partial shading.
- Complex control algorithms required for dynamic MPP tracking.
- DC/DC converters enable control of array operating point.



Proposed System

- 6 independent synchronous boost MPPT channels.
- Per-array tracking to minimise partial shading losses.
- Dual-output architecture supporting LV and HV buses.
- Automatic HV charging transition when LV is at capacity.
- STM32-based control with high-resolution PWM.
- Web-based UI for experimental algorithm deployment, telemetry monitoring, and array validation.
- Support for FD-CAN and serial communication.
- Designed for high reliability and seamless integration.

Maximises energy extraction under dynamic operating conditions while maintaining reliability within strict system constraints.



SR8 - Sunswift 8
ADR - Australian Design Rules
LV - Low Voltage
HV - High Voltage

SC - Short Circuit
OC - Open Circuit
MPPT - Maximum Power Point Tracker
MPP - Maximum Power Point

PWM - Pulse Width Modulation
CAN - Controller Area Network
COTS - Commercial Off-The-Shelf
GaN - Gallium Nitride